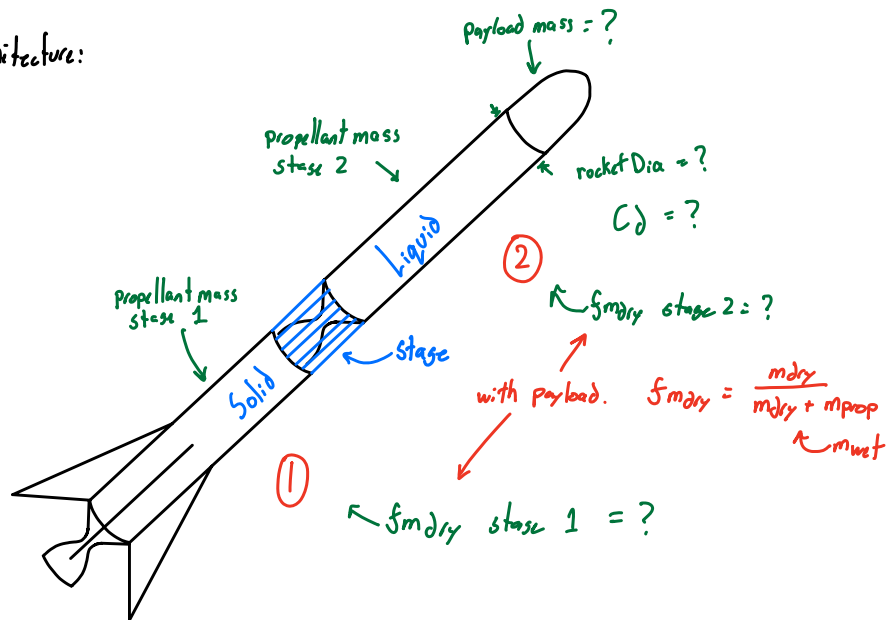


Launch Vehicle Optimization:

Problem Summary:

Design 2 stage (solid + liquid) sounding rocket - esgue launch. minimize propellant.
Think Aerobus but it can get to orbit.

System Architecture:



User Defined Variables:

Stage 1:

A_{throat} , A_{exit}

T_0 , P_0 out from CC

γ , R (from Stoichiometric Ratio/chem)

Stage 1:

Pure Gravity Turn: No Lift (Depending on Complexity / Results, may add vectors)
Strong Passive Stability

Initial Conditions: $\vec{r} = [r_0; 0]$ $\vec{v} = [0; 0]$ $\dot{m} = \text{constant}$ $m = M$ $\gamma_0 = \text{Launch Angle}$
(can reuse Midterm Code)

No need to consider relative position of c_g vs c_p

User Inputs:

Thrust/Time Curve: $[T, t]$ values for solid Motor

Physical Dimensions of Rocket

if there is time, can be generated from Nozzle
need Altitude correction

Launch Coordinate System:
 $\hat{h}$ out of page



Coast Phase:

Need dynamics?

Coast until above Atmosphere

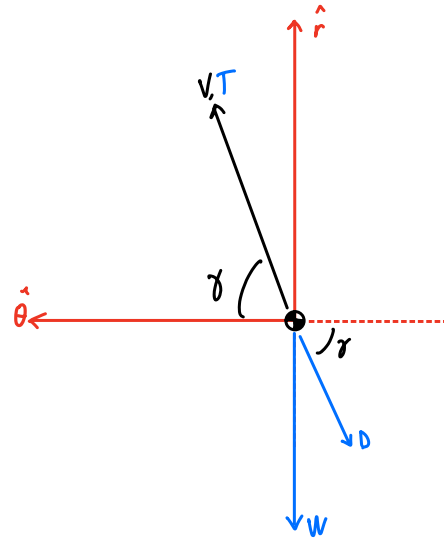
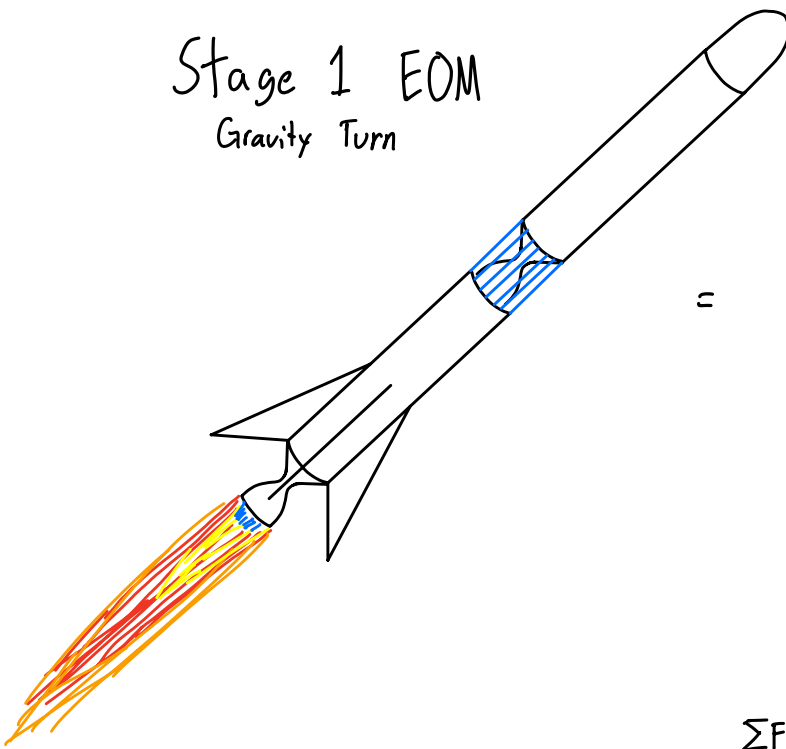
Stage 2:

Above Atmo

Derive EOM

Stage 1 EOM

Gravity Turn



$$\Sigma F = m \ddot{a}:$$

$$m \ddot{a}_r = T \sin(\gamma) - D \sin(\gamma) - W$$

$$m \ddot{a}_\theta = T \cos(\gamma) - D \cos(\gamma)$$

Where:

$$T = \dot{m} V_e + (P_e - P_{atm}) A_e$$

$$W = m M_e G / r^2$$

$$D = \frac{1}{2} \rho V^2 C_d A_{ref}$$

So:

$$\ddot{a}_r = \frac{\sin(\gamma)}{m} [\dot{m} V_e + (P_e - P_{atm}) A_e] - \frac{\sin(\gamma)}{2m} \rho V^2 C_d A_{ref} - \frac{G M_e}{r^2}$$

$$\ddot{a}_\theta = \frac{\cos(\gamma)}{m} [\dot{m} V_e + (P_e - P_{atm}) A_e] - \frac{\cos(\gamma)}{2m} \rho V^2 C_d A_{ref}$$

What are $\ddot{r}$ & $\ddot{\theta}$?

From 562:

$$\ddot{a}_r = \ddot{r} - r \dot{\theta}^2$$

$$\ddot{a}_\theta = r \ddot{\theta} + 2 \dot{r} \dot{\theta}$$

So:

$$\ddot{r} = \ddot{a}_r + r \dot{\theta}^2$$

$$\ddot{\theta} = \frac{\ddot{a}_\theta - 2 \dot{r} \dot{\theta}}{r}$$

ODE:

$$\ddot{r} = \frac{\sin(\gamma)}{m} [\dot{m} V_e + (P_e - P_{atm}) A_e] - \frac{\sin(\gamma)}{2m} \rho V^2 C_d A_{ref} - \frac{G M_e}{r^2} + r \dot{\theta}^2$$

$$\ddot{\theta} = \frac{\cos(\gamma)}{r m} [\dot{m} V_e + (P_e - P_{atm}) A_e] - \frac{\cos(\gamma)}{2 m r} \rho V^2 C_d A_{ref} - \frac{2 \dot{r} \dot{\theta}}{r}$$

Or:

$$\ddot{r} = \frac{\sin \gamma}{m} T - \frac{\sin \gamma}{m} D - g + r \dot{\theta}^2$$

$$\ddot{\theta} = \frac{\cos \gamma}{r m} T - \frac{\cos \gamma}{m r} D - \frac{2 \dot{r} \dot{\theta}}{r}$$

Non Constant Variables:

γ - flight path angle

m - rocket Mass

V_e - exhaust Velocity

P_e - exhaust Static Pressure

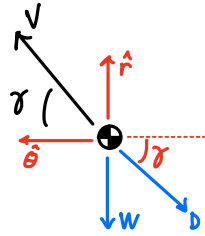
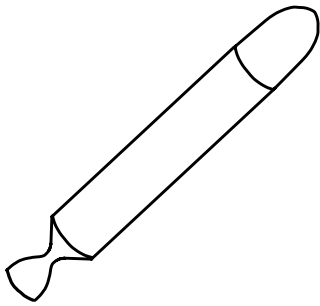
P_{atm} - free Stream Pressure

ρ - free stream density

V - rocket speed

r - rocket distance from COM of Earth

Coast Phase



$$\Sigma F = ma$$

$$m\bar{a}_r = -D \sin(\gamma) - W$$

$$m\bar{a}_\theta = -D \cos(\gamma)$$

$$W = m M_e G / r^2$$

$$D = \frac{1}{2} \rho V^2 C_D A_{ref}$$

What are $\ddot{r}$ & $\ddot{\theta}$?

From 562:

$$\bar{a}_r = \ddot{r} - r\dot{\theta}^2$$

$$\bar{a}_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

So:

$$\ddot{r} = \bar{a}_r + r\dot{\theta}^2$$

$$\ddot{\theta} = \frac{\bar{a}_\theta - 2\dot{r}\dot{\theta}}{r}$$

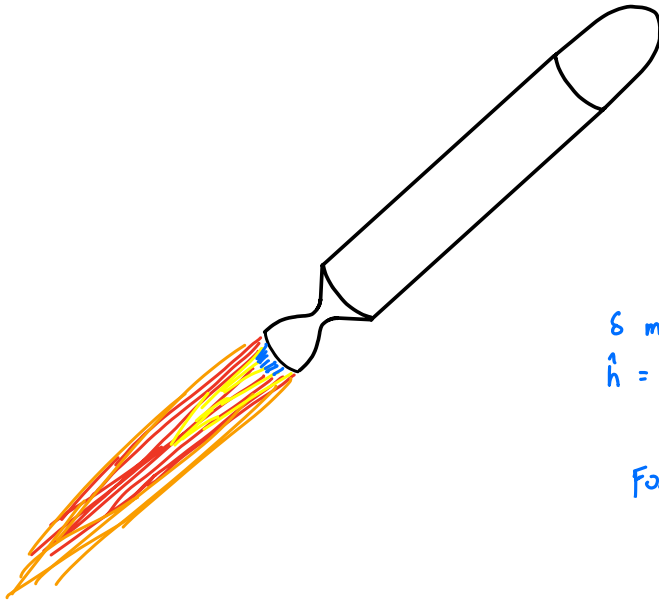
ODE:

$$\ddot{r} = -\frac{D}{m} \sin(\gamma) - g + r\dot{\theta}^2$$

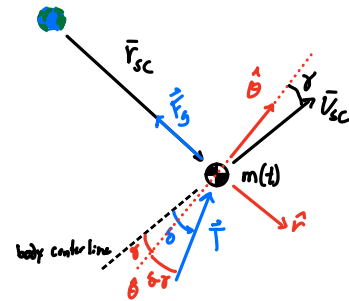
$$\ddot{\theta} = -\frac{D}{rm} \cos \gamma - 2\dot{r}\dot{\theta}/r$$

Stage 2 EOM Derivation

System Diagram



=



δ measured around

$\hat{h} = \hat{r} \times \hat{\theta}$ per RHR

Forces:

$$\vec{F}_g = -\frac{m \cdot M_E \cdot G}{r^3} \vec{r}$$

control vector
 $\delta = \vec{u}(t)$

$$\vec{T} = [\dot{m}V_e + (\cancel{P_e} - P_e)A_e] \cdot [-\sin(\delta - \gamma), \cos(\delta - \gamma)]$$

no Atmo Assumption

Net:

$$m \bar{a}_r = -\frac{m M_E G}{r^2} - \sin(\delta - \gamma) [\dot{m}V_e + P_e A_e]$$

$$m \bar{a}_\theta = \cos(\delta - \gamma) [\dot{m}V_e + P_e A_e]$$

What are $\ddot{r}$ & $\ddot{\theta}$?

From 562:

$$\bar{a}_r = \ddot{r} - r\dot{\theta}^2$$

$$\bar{a}_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

So:

$$\ddot{r} = \bar{a}_r + r\dot{\theta}^2$$

$$\ddot{\theta} = \frac{\bar{a}_\theta - 2\dot{r}\dot{\theta}}{r}$$

ODE:

$$\text{Stage 2 ODE} \begin{cases} \ddot{r} = -\frac{M_E G}{r^2} - \frac{\sin(\delta - \gamma)}{m} [\dot{m}V_e + P_e A_e] + r\dot{\theta}^2 = -g - \frac{\sin(\delta - \gamma)}{m} T + r\dot{\theta}^2 \\ \ddot{\theta} = \frac{\cos(\delta - \gamma)}{m r} [\dot{m}V_e + P_e A_e] - \frac{2\dot{r}\dot{\theta}}{r} = \frac{\cos(\delta - \gamma)}{m r} T - \frac{2\dot{r}\dot{\theta}}{r} \end{cases}$$

Summary:

Stage 1:

$$\ddot{r} = \frac{\sin \delta}{m} T - \frac{\sin \delta}{m} D - g + r\dot{\theta}^2$$

$$\ddot{\theta} = \frac{\cos \delta}{mr} T - \frac{\cos \delta}{mr} D - \frac{2\dot{r}\dot{\theta}}{r}$$

Coast:

$$\ddot{r} = -\frac{D}{m} \sin(\delta) - g + r\dot{\theta}^2$$

$$\ddot{\theta} = -\frac{D}{mr} \cos \delta - 2\dot{r}\dot{\theta}/r$$

Stage 2:

$$\ddot{r} = -g - \frac{\sin(\delta-\gamma)}{m} T + r\dot{\theta}^2$$

$$\ddot{\theta} = \frac{\cos(\delta-\gamma)}{mr} T - \frac{2\dot{r}\dot{\theta}}{r}$$

General:

$$\ddot{r} = \frac{-\sin(\delta-\gamma)}{m} T - \frac{\sin(\gamma)}{m} D - g + r\dot{\theta}^2$$

$\delta=0$ Case 1: ✓✓
 $\delta=T=0$ Case 2: ✓✓
 $D=0$ Case 3: ✓✓

$$\ddot{\theta} = \frac{\cos(\delta-\gamma)}{mr} T - \frac{\cos(\gamma)}{mr} D - \frac{2\dot{r}\dot{\theta}}{r}$$

Where:

↙ Can't bc machine precision

State:

$$\begin{bmatrix} r \\ \theta \\ \dot{r} \\ \dot{\theta} \\ m \end{bmatrix}$$

$$T = \dot{m} V_e + (P_e - P_{atm}) A_e$$

$$\hookrightarrow V_e = M_e \sqrt{\gamma R T_e}$$

↪ Area Mach Relation

$$\hookrightarrow P_{atm} \rightarrow \text{lookup table } P_{atm}(r-r_E)$$

$$\hookrightarrow P_e = P_0 \left(\frac{T_e}{T_0} \right)^{\gamma/(\gamma-1)}$$

$$\hookrightarrow T_e = T_0 \left(1 + \frac{\gamma-1}{2} M_e^2 \right)^{-1}$$

$$\hookrightarrow \dot{m} \rightarrow \text{from isentropic } E_g, \text{ assumed to be constant}$$

$$\hookrightarrow A_e \rightarrow \text{user input}$$

$$W = m \bar{M}_E G / r^2$$

$$\hookrightarrow \text{Mass of Earth, } G \rightarrow \text{constants}$$

$$\hookrightarrow m \text{ \& } r, \text{ state variables}$$

$$D = \frac{1}{2} \rho V^2 C_D A_{nf}$$

$$\hookrightarrow V = \sqrt{\dot{r}^2 + r^2 \dot{\theta}^2}$$

$$\hookrightarrow C_D, A_{nf} \rightarrow \text{user inputs}$$

$$\hookrightarrow \rho \rightarrow \text{table lookup } \rho(h)$$

Case Switch Conditions

$$1 \rightarrow 2: m \leq M_i - M_{\text{solid prop}}$$

$$2 \rightarrow 3: r \geq r_{\text{endAtmo}}$$

Open parts for expansion:

~~m has no function file~~

No $\hookrightarrow P_0, T_0$ may change int

T, W, D function files should

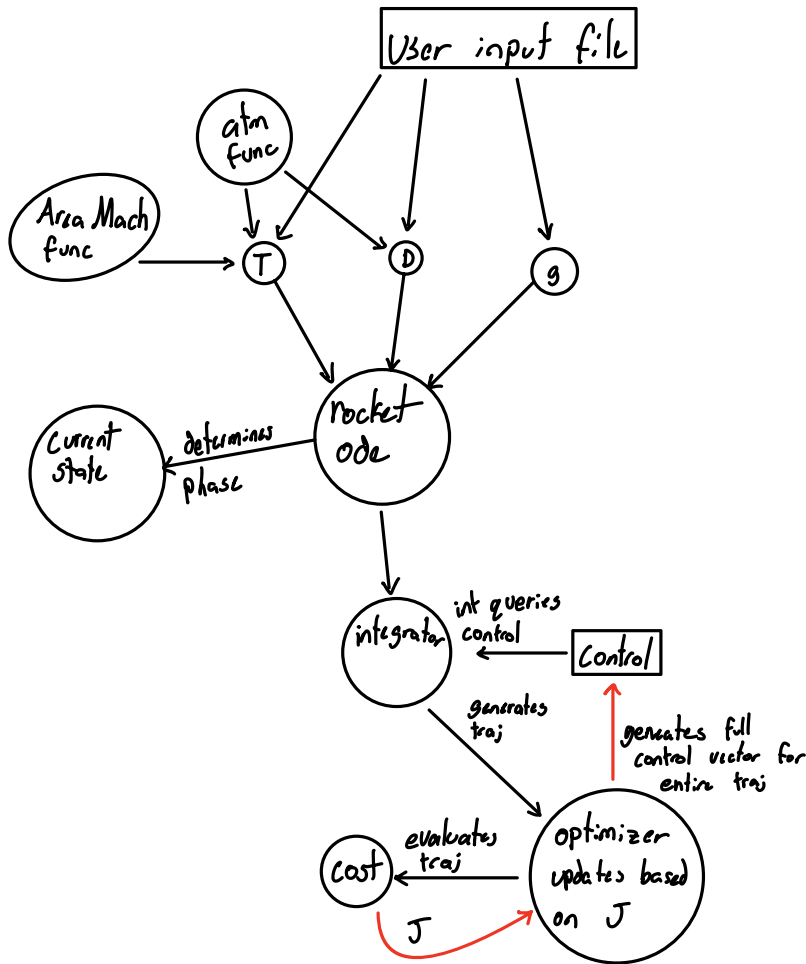
D $\neq 0$ + $\delta \neq 0$ case be added

T esp. should prop aspects want to be added

We assume constant chamber pressure

decouple

Architecture:



A → B
 B calls/queries A
 text file
 → sends value

ODE/ Integrator / main

ODE: stage number in params
 Vary D, T, m, etc by stage
 one set of EOM

$$\text{state} = [r, \theta, \dot{r}, v_\theta, m]$$

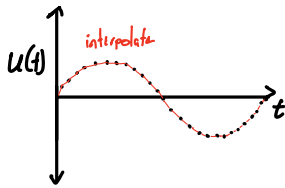
$$\partial X = [\dot{r}, v_\theta/r, \ddot{r}, \dot{v}_\theta, -\dot{m}]$$

integrator watches for staging events
 and then stops & restarts integration.
 that way we can 1: drop engine mass
 and 2) easier to debug

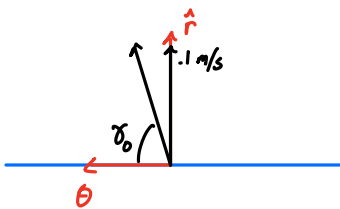
optimizer picks

integrator time steps vs control time steps.
 ode45 uses variable time steps
 optimizer returns control values at nodes

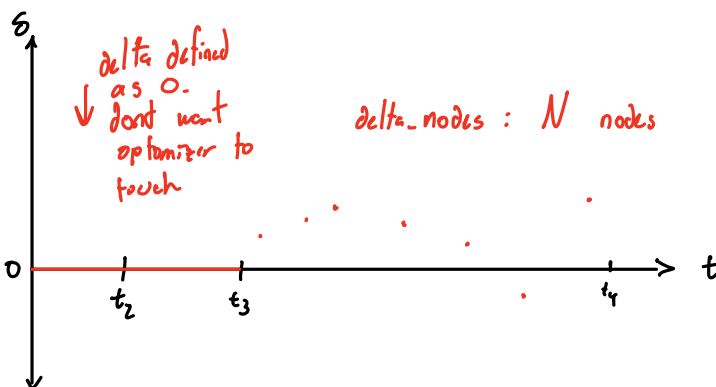
Need function to take user input file
 and output params structure ✓



Initial Condition



$$X_0 = \begin{cases} r_0 = \text{radius Earth} \\ \theta_0 = 0 \\ \dot{r} = 0.1 \\ \dot{\theta} = \text{selected to meet } \theta_0 \\ m = M_i \end{cases}$$



optimization vector:

$$x = [\text{solidPropMass}, \text{liquidPropMass}, \text{launch Angle}, \text{delta}_1, \text{delta}_2, \dots, \text{delta}_N]$$

We know $t_4 - t_3 = \text{burn time of liquid}$
 this can be computed analytically

In rocketIntegrator, add tstage3 in transition 2-3
 In Ode call deltaInterp function @ $t - t_{\text{stage3}}$
 t_{nodes} & $\text{delta}_{\text{nodes}}$ are normalized to $[0, t_{\text{burns}}]$

Constraints:

Inequalities

$$c_1 = (a_{\text{final}} - a_{\text{target}}) - a_{\text{margin}} \leq 0$$

$$c_2 = -(a_{\text{final}} - a_{\text{target}}) - a_{\text{margin}} \leq 0$$

$$c_3 = (e_{\text{final}} - e_{\text{target}}) - e_{\text{margin}} \leq 0$$

$$c_4 = -(e_{\text{final}} - e_{\text{target}}) - e_{\text{margin}} \leq 0$$

Optimizer, Cont.

Integrator will generally fail to establish an orbit.
need to catch these & attach large penalty value

Variable Bounds

mass: $5 \times \text{guess} - 0.01 \times \text{guess}$

launch angle: $[0.785, 1.5708] \text{ rad}$ $[45^\circ - 90^\circ]$

δ : $[-\text{maxThrustAngle}, +\text{maxThrustAngle}]$

Need fairly close IG

user input value: generate IG_bool = 1

function to Generate IG

Guess solid prop mass based off historical values relative to payload size?

Integrate to start of phase 3, then guess ΔV from Tsiolkovsky's Eq

guess $\delta = 0$ (so always radial?)

IG algorithm

First:

Guess masses as payload multiples

K_{solid} , K_{liquid} $\leftarrow$ historical values?

loop: till m_{liquid} converges:

$M_{\text{li_guess}} \leftarrow$ from prop masses

run integrator to stage 3

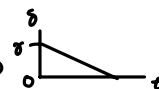
compute required ΔV from state

Tsiolkovsky $\rightarrow$ new liquid mass estimate

update liquid prop mass & iterate until no change

Assemble & return guesses

δ will be linear decrease from δ @ start of 3 to zero



User Input File

% Target Orbit Info (2D)

a : orbit semi major axis
e : orbit eccentricity

m

% Planet parameters

radius Earth : radius of Earth
mass Earth : mass of Earth
Grav Constant : gravitational Constant G
atmo End Percnt : Atmosphere is considered to end when it reaches
this percentage of the sea level value.

m

kg

$\text{m}^3/\text{kg}\cdot\text{s}^2$

% Engine Parameters

% Stage 1:

totalTemp1 : combustion Chamber total temp (assumed constant)
totalPressure1 : combustion Chamber total pressure (assumed constant)
gamma1 : propellant mixture specific heat ratio
exitArea1 : Nozzle exit Area
throatArea1 : Nozzle throat Area
gasConstant1 : propellant mixture specific gas constant

K

Pa

m^2

m^2

$\text{J}/\text{kg}\cdot\text{K}$

% Stage 2:

totalTemp2 : combustion Chamber total temp (assumed constant)
totalPressure2 : combustion Chamber total pressure (assumed constant)
gamma2 : propellant mixture specific heat ratio
exitArea2 : Nozzle exit Area
throatArea2 : Nozzle throat Area
gasConstant2 : propellant mixture specific gas constant

K

Pa

m^2

m^2

$\text{J}/\text{kg}\cdot\text{K}$

% Rocket Parameters

dragCoefficient : Rocket Drag coefficient @ 0° Aof
referenceArea : Drag Reference Area
(Generally cross sectional area at widest point)
payloadMass : mass of payload. Includes payload structures
engineMass1 : mass of solid Motor
engineMass2 : mass of liquid engine
massFrac1 : % mass of solid propellant for structures
massFrac2 : % mass liquids (tanks, fuel lines, casings, lining, etc)

m^2

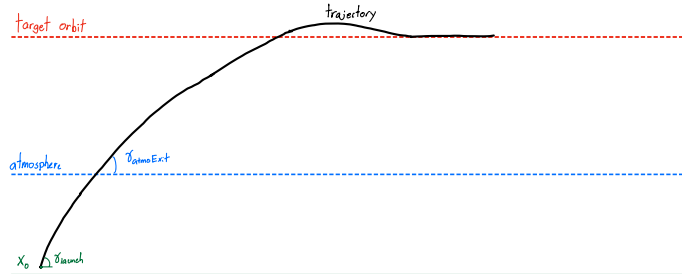
kg

kg

kg

% Initial Guess / Optimizer Setup

max Thrust Angle	:	max angle from body centerline of Thrust	rad
liquid Prop Mass IG	:	mass of L prop (S2) initial Guess	kg
solid Prop Mass IG	:	mass of S prop (S1) initial Guess	kg
launch Angle IG	:	initial flight path angle to start Gravity Turn	rad



Further refinement ideas:

Decision variable to start liquid burn when?

Lift + AoA

implement mission feasibility check:

$$I_{sp} \geq \frac{\Delta V}{\ln\left(\frac{1}{f_{int}}\right) g_0} \quad \text{where } f_{int} = \frac{m_{int}}{m_{int} + m_{prop}} \quad (\text{m}_{int} \text{ does not include payload})$$

add ΔV objective